



Adafruit bq25185 USB / DC / Solar Charger with 5V Boost Board

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<https://learn.adafruit.com/adafruit-bq25185-usb-dc-solar-charger-with-5v-boost-board>

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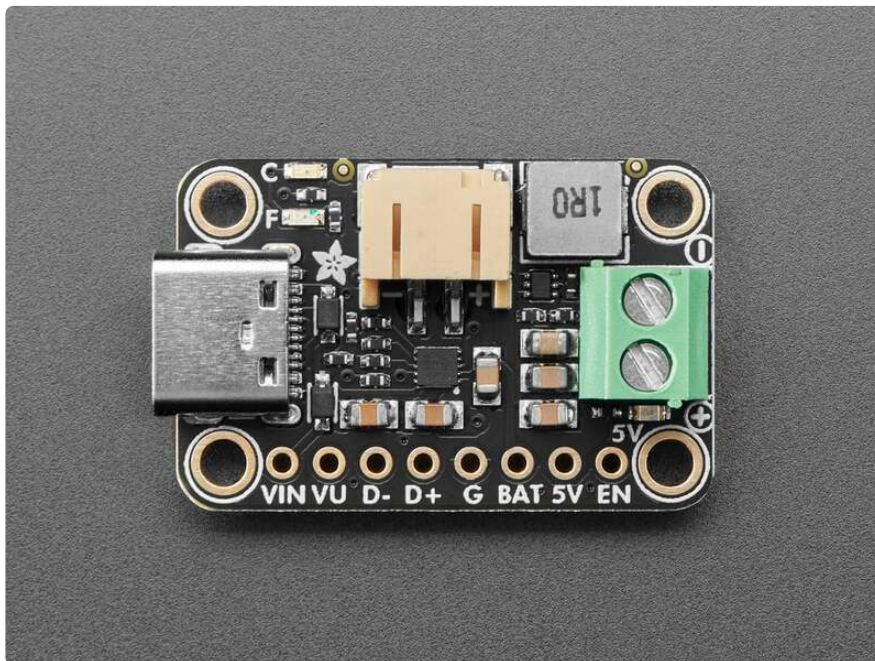
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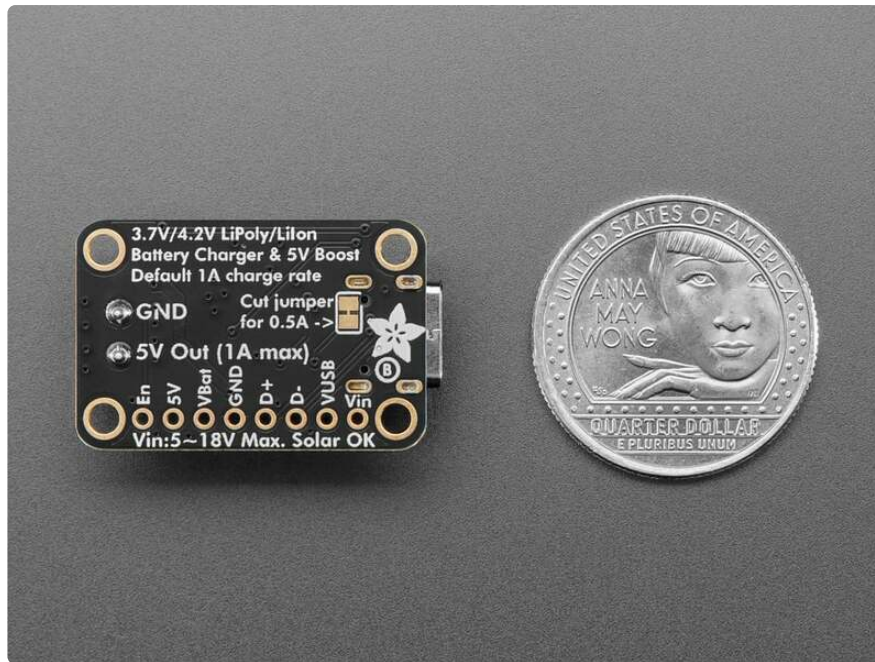
Overview



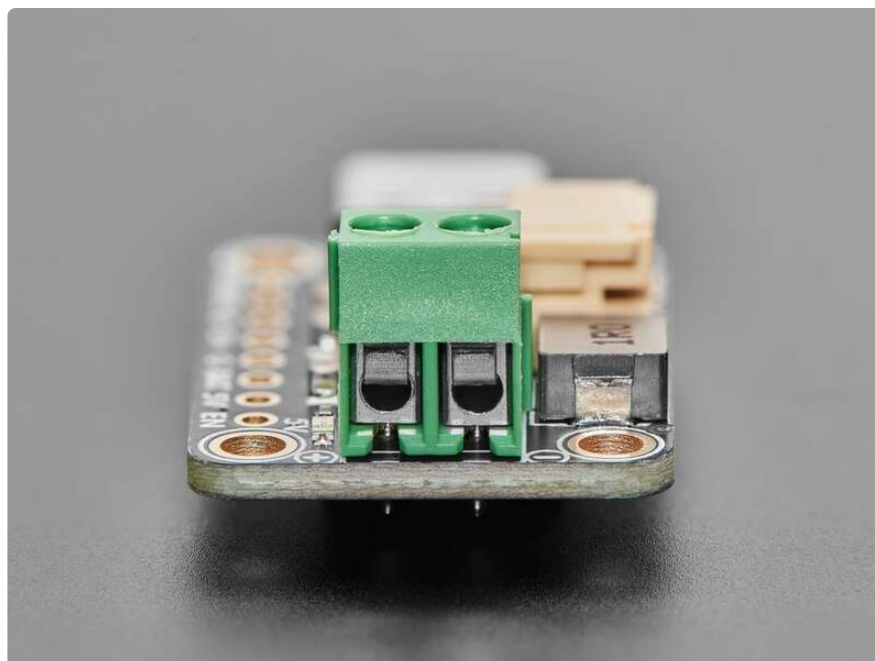
There is an unmodifiable 6 hour safety timeout on this chip. If the board is charging for 6 hours without any interruptions, charging will stop.



We're always on the look out for better ways to make projects portable: being able to charge your battery in the most convenient manner will let projects run no matter what power is available. Then we added a power supply chip with it to let you run your project without a separate board. The result is the **Adafruit bq25185 USB / DC / Solar Charger with 5V Boost Board!**

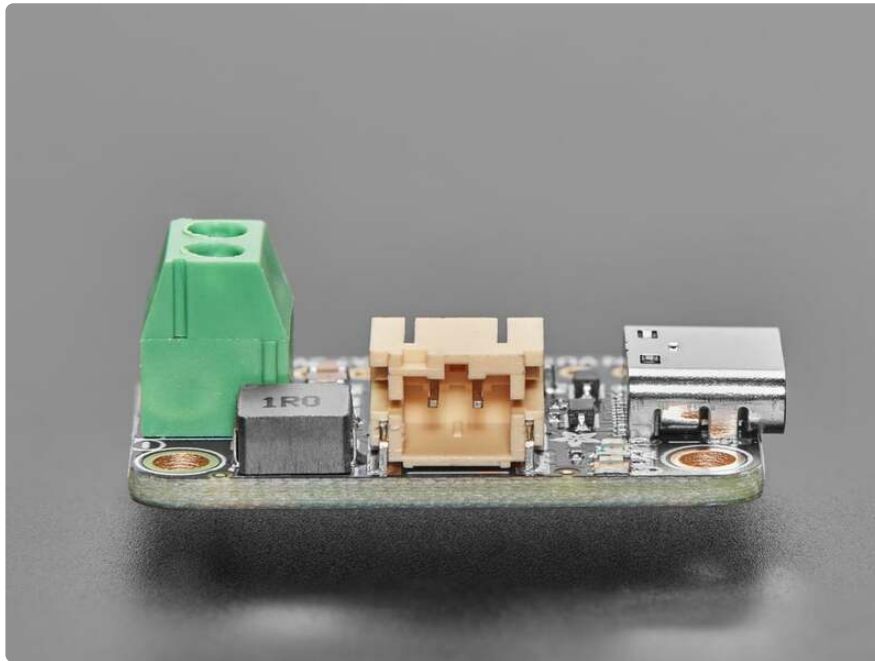


It uses the new [bq25185](https://adafru.it/1aau) (<https://adafru.it/1aau>), a nifty charger chip with a fairly high charge current, power path support, and the ability to charge from USB, DC or solar power. It's also a great value, so it's a good upgrade from [MCP73833](http://adafru.it/259) (<http://adafru.it/259>) or [MCP73831-based](http://adafru.it/4410) (<http://adafru.it/4410>) charger boards. The boost converter is the [TPS61023](http://adafru.it/4654) (<http://adafru.it/4654>), which will give up to 1 Amp output with good efficiency, to squeeze the most power possible out of your battery.



This board is meant to be everything you need to power your 5V electronics: simply connect a 500mAh or larger battery to the JST PH 2-pin port, then charge it when you

can from USB or DC/solar. At the other end is a terminal block which will provide the 5V output from the boost converter.



- **USB Type C** connector with 5.1k CC resistors so it will work with any computer or power supply to get 5V and up to 1A. Data lines available if needed, on the bottom breakout.
- **Separate DC or Solar Input** - Two pads on the edge can be used to connect a 5 ~ 18V power supply, which can be used instead of USB. If the input is a solar panel, the charging chip will adjust the current draw so that the voltage does not dip below the battery, thus optimizing the solar power input. No large capacitor needed to stabilize it, and you get near-MPPT capability without the cost and complexity of MPPT.
- **Default charge rate of 1A**, but you can cut the jumper on the back to set the rate to 500mA
- **Power Path to Load** - If the 5V load connector is drawing current while the USB / DC/Solar power is attached, it will default to drawing current from the charger and any left over current will go to the battery. That keeps your battery from constantly charging/discharging which will reduce the battery life.
- **Regulated 5V Output** - a boost converter will take the load output from the bq25185 and convert it to 5V at 1 Amp max load. Note the TP61023 cannot turn on/off high loads, it will stall over 200mA. That means its not going to like it if you have a high current load that instantly draws full current on power up, like a resistive dummy load. The vast number of microcontroller projects will start up

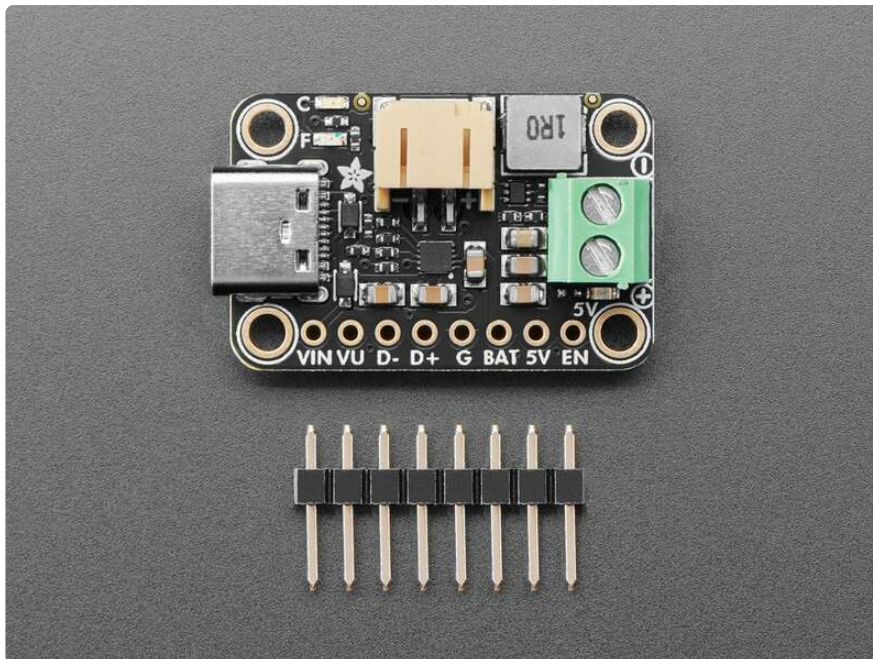
with <200mA and then a half second later will turn on high current items like LEDs, motors, wifi/radios, or displays so the restriction won't affect it.

- **Three Status LEDs** - Orange Charging LED, red Fault LED, and Green 5V output LED.
- **Enable pad** - to disable the 5V boost converter.
- Mounting holes!



This board is pretty much plug-and-play. Change the charging jumper if you like, then connect your battery to the BATT port, and the 5V output goes to your circuit. You can monitor the voltage on the battery via the secondary pads on the bottom edge, if necessary.

To use with solar, [pick up a 5 to 7V solar panel \(https://adafruit.it/1aav\)](https://adafruit.it/1aav) (using a higher voltage panel will only lose the extra voltage as heat so there's no benefit to going over 7V), and either cut the connector off to wire it directly, or [use a 2.1mm adapter cable \(http://adafruit.it/2788\)](http://adafruit.it/2788) plus a [2.1mm terminal block \(http://adafruit.it/368\)](http://adafruit.it/368) to get two wires for the DC Input port.

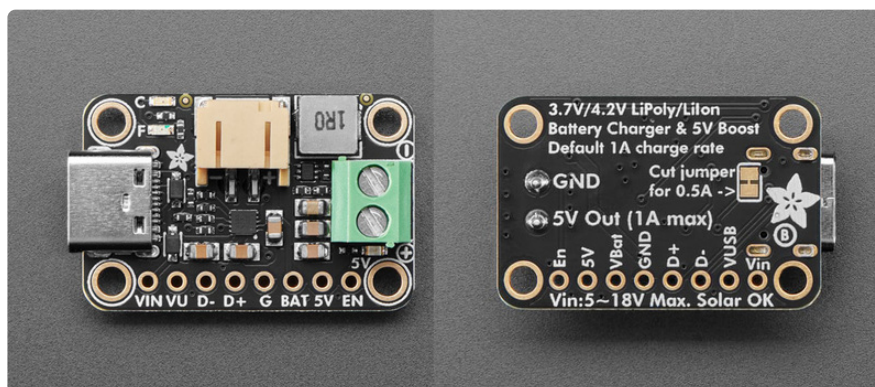


If you need a board with higher charge current or a DC plug already on-board, [check out the bq24074](http://adafru.it/4755) which has up to 1.5A charge rate and an on-board 2.1mm DC jack (<http://adafru.it/4755>). It doesn't have a boost converter but you could wire up the [TPS61023](http://adafru.it/4654) (<http://adafru.it/4654>) separately.

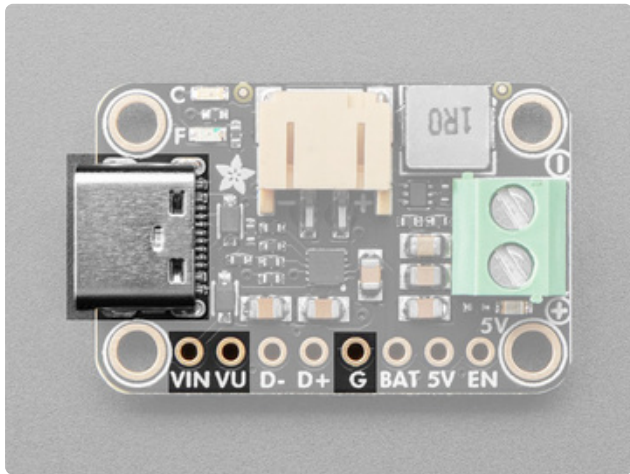
Pinouts



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Power Inputs



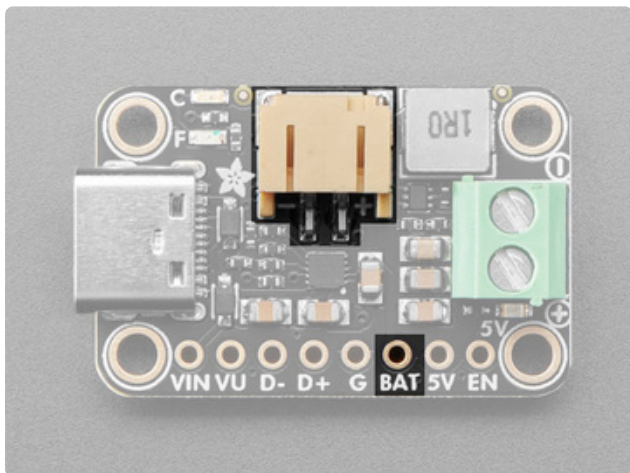
USB Type C - On the edge of the board is the **USB Type C** connector. It has 5.1k CC resistors so it will work with any computer or power supply to get 5V and up to 1A.

VIN - This is the solar panel or DC power input to the bq25185. Connect a 5 to 18V power supply. If the input is a solar panel, the charging chip will adjust the current draw so that the voltage does not dip below the battery, thus optimizing the solar power input.

VU - This is the power input from the USB connector.

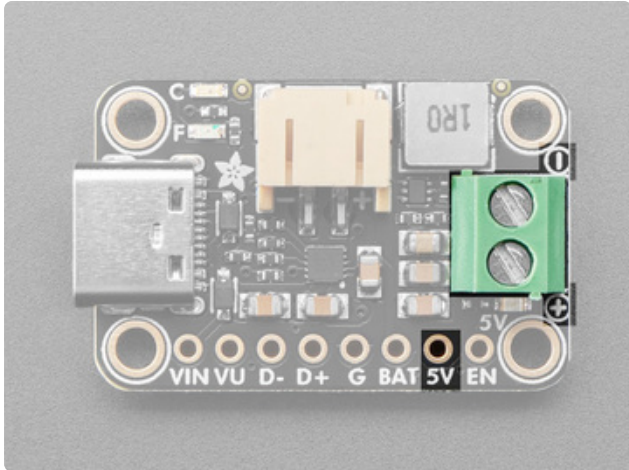
G - common ground.

Battery Output



At the top of the board is the JST battery connector. By default, the output is 3.7V nominal / 4.2V max battery chemistry/ voltage for all modern 1-cell LiPoly or Lilon batteries with a default charge rate of 1A. This output is available on the bottom of the board as well, labeled **BAT** on the board silk. You can use this secondary output to monitor the voltage on the battery with a multimeter.

Terminal Block Output

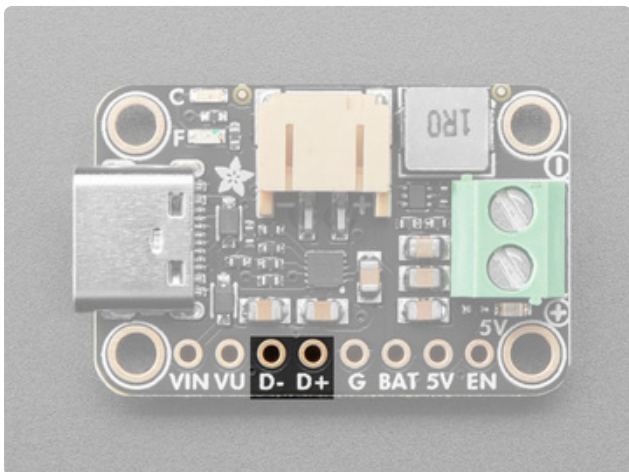


- - This is common ground.

+ - This is the regulated 5V output from the separate buck converter. It takes the load output from the bq25185 and switches it to 5V at 1 Amp max load. This output is available on the bottom of the board as well, labeled **5V** on the board silk.

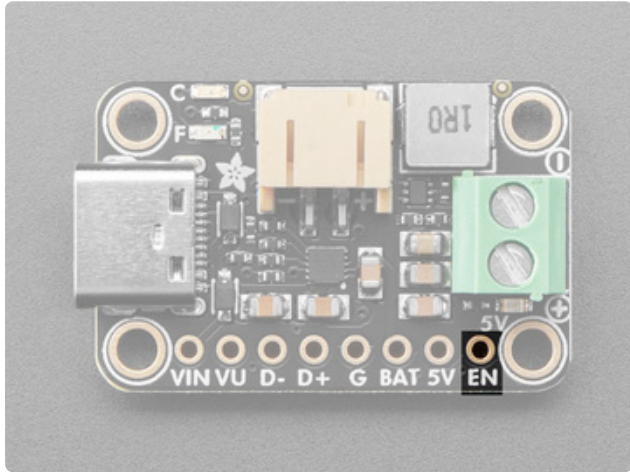
Note the TP61023 cannot turn on/off high loads; it will stall over 200mA. That means it's not going to like it if you have a high current load that instantly draws full current on power up, like a resistive dummy load. The vast number of microcontroller projects will start up with <200mA and then a half second later will turn on high current items like LEDs, motors, WiFi/radios, or displays so the restriction won't affect it.

USB Data Pins



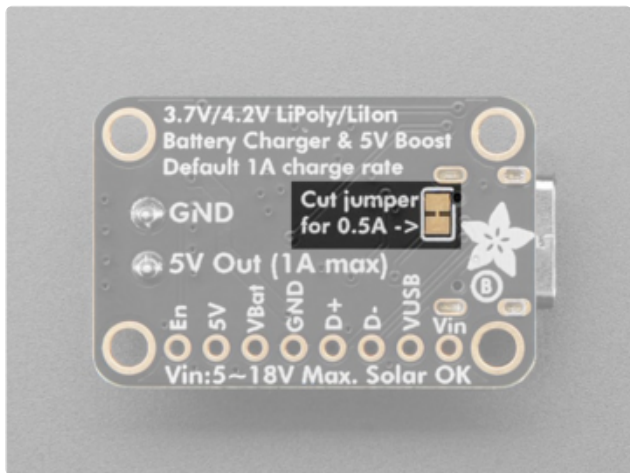
D+ and **D-** - USB positive and negative data lines

Enable Pin



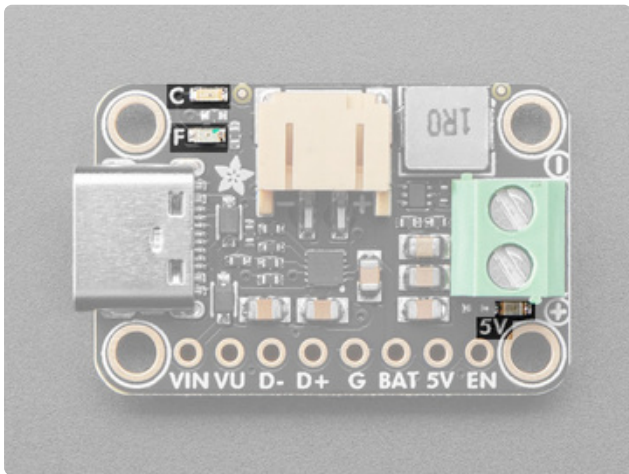
EN - This is the enable pad for the 5V output from the boost converter. Bring this pin low to disable the 5V output.

Charge Rate Jumper



On the back of the board is the **charge rate jumper**. You can cut this jumper to change the charge rate from 1A to 500mA.

Status LEDs



5V Output LED - In the bottom right corner of the board, below the terminal block, is the 5V output LED. It is the green LED. It will be lit if the 5V output is enabled.

Charge LED - In the top left corner of the board is the charge status LED. It is the orange LED. It is labeled C on the board silk. It will be lit while a battery is charging.

Fault LED - Directly below the charge status LED is the fault status LED. It is the red LED. It is labeled F on the board silk. It will be lit if the board experiences a fault condition. The bq25185 has integrated fault protection for a few conditions:

- Input overvoltage protection
- Battery undervoltage protection
- Battery short protection
- Battery overcurrent protection
- Input current limit protection
- Thermal regulation and thermal shutdown
- Battery thermal fault protection
- Safety timer fault
- ISET and ILIM/VSET pin short/open protection.

Downloads



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Files

- [bq25185 Datasheet \(https://adafru.it/1aaw\)](https://adafru.it/1aaw)
- [EagleCAD PCB Files on GitHub \(https://adafru.it/1aaV\)](https://adafru.it/1aaV)
- [Fritzing object in the Adafruit Fritzing Library \(https://adafru.it/1aaW\)](https://adafru.it/1aaW)

Schematic and Fab Print

